Lecture 8

Techniques of Differentiation

Parametric Differentiation:

Example: If $x = a\cos\theta$ and $y = b\sin\theta$ where $0 \le \theta \le 2\pi$, find $\frac{dy}{dx}$.

Solution: $\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{b\cos\theta}{-a\sin\theta} = -\frac{b}{a}\cot\theta$.

Example: If $x = a(\theta + \sin\theta)$, $y = a(1 - \cos\theta)$, find $\frac{dy}{dx}$.

Solution: $\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{a(0+\sin\theta)}{a(1+\cos\theta)} = \frac{\sin\theta}{1+\cos\theta}$

Example: If $x = \sin^2\theta$, $y = \tan\theta$, find $\frac{dy}{dx}$.

Solution: Here, $x = \sin^2\theta = (\sin\theta)^2$

So, $\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{\sec^2\theta}{2\sin\theta\cos\theta} = \frac{\sec^2\theta}{\sin 2\theta}$

H.W. Practice Sheet 5

Differentiating $f(x)$ with respect to $g(x)$:

$y \quad = f(x)$
$z \quad = g(x)$

$$\frac{dy}{dz} = \frac{\frac{dy}{dx}}{\frac{dz}{dx}}.$$

Example: Differentiate $x^{\sin^{-1}x}$ with respect to $\sin^{-1}x$.

Solution: Let, $y \quad = x^{\sin^{-1}x}$, $\quad z = \sin^{-1}x$

Now, $y \quad = x^{\sin^{-1}x}$

⇨ $\ln y \quad = \ln\left(x^{\sin^{-1}x}\right)$

⇨ $\ln y = \sin^{-1}x \cdot \ln x$

Differentiating both sides with respect to x,

$$\frac{d}{dx}(\ln y) = \frac{d}{dx}(\sin^{-1}x \cdot \ln x)$$

$$\Rightarrow \frac{1}{y}\frac{dy}{dx} = \frac{\sin^{-1}x}{x} + \frac{\ln x}{\sqrt{1-x^2}}$$

$$\Rightarrow \frac{dy}{dx} = y\left[\frac{\sin^{-1}x}{x} + \frac{\ln x}{\sqrt{1-x^2}}\right]$$

$$\Rightarrow \frac{dy}{dx} = x^{\sin^{-1}x}\left[\frac{\sin^{-1}x}{x} + \frac{\ln x}{\sqrt{1-x^2}}\right]$$

Again,

$$\frac{dz}{dx} = \frac{d}{dx}(\sin^{-1}x) = \frac{1}{\sqrt{1-x^2}}$$

So,

$$\frac{dy}{dz} = \frac{\frac{dy}{dx}}{\frac{dz}{dx}} = \frac{x^{\sin^{-1}x}\left[\frac{\sin^{-1}x}{x} + \frac{\ln x}{\sqrt{1-x^2}}\right]}{\frac{1}{\sqrt{1-x^2}}}$$

H.W. Practice Sheet 6 (i)

Example: Find the differential coefficient of $\ln\sqrt{\frac{1+\sin x}{1-\sin x}}$.

Solution: let, $y = \ln\sqrt{\frac{1+\sin x}{1-\sin x}}$

$$\Rightarrow y = \ln\left(\frac{1+\sin x}{1-\sin x}\right)^{\frac{1}{2}}$$

$$\Rightarrow y = \frac{1}{2}\ln\left(\frac{1+\sin x}{1-\sin x}\right)$$

$$\Rightarrow \frac{dy}{dx} = \frac{d}{dx}\left[\frac{1}{2}\ln\left(\frac{1+\sin x}{1-\sin x}\right)\right]$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{2}\left[\frac{d}{dx}[\ln(1+\sin x) - \ln(1-\sin x)]\right]$$

$$= \frac{1}{2}\left[\frac{1}{1+\sin x}\cdot\cos x - \frac{1}{1-\sin x}\cdot(-\cos x)\right]$$

$$\ln(xy) = \ln x + \ln y$$

$$\ln\left(\frac{x}{y}\right) = \ln x - \ln y$$

$$= \frac{1}{2}\left[\frac{\cos x}{1+\sin x} + \frac{\cos x}{1-\sin x}\right]$$

$$= \frac{1}{2}\left[\frac{\cos x(1-\sin x) + \cos x(1+\sin x)}{(1+\sin x)(1-\sin x)}\right]$$

$$= \frac{1}{2}\left[\frac{\cos x - \cos x \cdot \sin x + \cos x + \cos x \cdot \sin x}{1-\sin^2 x}\right]$$

$$= \frac{1}{2} \times \frac{2\cos x}{1-\sin^2 x}$$

$$= \frac{1}{2} \times \frac{2\cos x}{\cos^2 x}$$

$$= \frac{1}{\cos x}$$

$$= \sec x$$

So, the differential coefficient is $\sec x$.

H.W. Practice Sheet 7